

Workflow Audit Table

#	Recurring Task	Classification	Rationale
1	Planning my weekly study schedule for ML, Python, and numerical methods	Collaborate with AI	AI can organize topics and create realistic schedules, but I decide priorities based on deadlines.
2	Solving advanced mathematics proofs and derivations	Just Me	I need to develop my own mathematical reasoning and cannot rely on AI for learning core concepts.
3	Writing Python code for assignments and internship tasks	Collaborate with AI	AI helps generate and debug code, but I must understand and verify every solution.
4	Debugging Python errors	Delegate to AI with review	AI usually identifies syntax and logic errors quickly, but I test every fix before using it.
5	Preparing lesson plans for tutoring students	Delegate to AI with review	AI saves time creating structured lesson plans, while I adapt them to each student's level.
6	Writing professional emails to professors, supervisors, and recruiters	Delegate to AI with review	AI improves clarity and professionalism, but I personalize every message before sending.
7	Researching PhD scholarships and university requirements	Collaborate with AI	AI helps summarize requirements, but I verify information on official university websites.

8	Reviewing research papers related to numerical analysis and machine learning	Collaborate with AI	AI summarizes papers and explains concepts, but I read important sections myself.
9	Creating LinkedIn posts and networking messages	Delegate to AI with review	AI improves writing quality while I ensure authenticity and accuracy.
10	Managing internship notes and documenting weekly progress	Fully automate	AI can organize notes, generate summaries, and maintain documentation consistently.
11	Preparing interview questions and educational material for tutoring	Delegate to AI with review	AI generates diverse practice questions, while I check correctness and difficulty level.
12	Checking scholarship eligibility and comparing universities	Collaborate with AI	AI helps compare programs efficiently, but the final decision requires my judgment.
13	Making final decisions about my PhD applications and career path	Just Me	These decisions involve personal goals, finances, and long-term priorities that only I can determine.

To better understand how AI can support my academic and professional work, I analyzed the recurring tasks I perform each week. Each task was classified according to the level of AI involvement that is appropriate, ranging from tasks that require my own judgment to tasks that can be automated.

Three Target Tasks

Target Task	Success Definition
Research paper analysis	Summarize the paper accurately, identify the methodology, key findings, and limitations, and connect it to my research interests in under one page.
Python programming and debugging	Produce code that runs without errors, is well commented, follows good coding practices, and that I fully understand before using.
Professional academic communication	Draft clear, personalized emails or LinkedIn messages that are grammatically correct, concise, and appropriate for the recipient.

Claude Project

I created a Claude Project titled "**PhD Research & AI Learning Assistant.**" The project includes custom instructions describing my academic background, learning goals, preferred response style, and expectations for AI collaboration. This configuration allows Claude to provide more consistent and context-aware support for my PhD preparation, AI learning, programming, and academic writing.

PhD Research & AI Learning Assistant / Preparing for PhD in applied mathematics and machine learning ▾

I am an MS Mathematics graduate specializing in numerical analysis and scientific computing. I am currently preparing applications for funded PhD positions while learning Python, machine learning, and artificial intelligence through independent study and an AI internship. I also teach mathematics to IB, A-Level, AP, and IGCSE students. My goal is to combine applied mathematics, numerical methods, and machine learning in future research. Please respond in a professional, clear, and educational tone, explaining concepts step by step and emphasizing understanding over simply providing answers. When helping with coding, explain the logic behind the solution, suggest best practices, and help me debug errors rather than only providing corrected code. When assisting with research, summarize papers accurately, highlight key methodologies and findings, identify limitations, and relate them to my research interests. For academic writing, help me draft professional emails, research summaries, LinkedIn messages, and application materials while maintaining a formal and concise style. Whenever appropriate, encourage critical thinking, point out common mistakes, distinguish facts from opinions, and recommend verifying important information through reliable or official sources. My primary goals are to strengthen my programming and AI skills, improve my research abilities, publish high-quality academic work, and secure a fully funded PhD position.

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Reflection

Completing this workflow audit helped me better understand where AI can genuinely improve my productivity and where my own judgment remains essential. I realized that AI is most effective as a collaborator for tasks such as programming, research, and academic writing, while decisions involving learning, ethics, and long-term career planning should remain my responsibility. This exercise has encouraged me to adopt AI more intentionally by using it to enhance efficiency without replacing critical thinking or personal accountability.